

All theorem/lemma/proposition numbering follows that of Chapter 5 of Puterman & Chao.

Lecture-08 [21 August 2026]

Throughout, we will assume $\mathcal{S}, \mathcal{A}$ finite

In the previous lecture, we saw that for any $\phi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$, the value function $v_{\infty, \lambda}^{\phi}$ is the unique solution to the equation

$$\underline{v} = r_{\phi} + \lambda P_{\phi} \underline{v} \quad (\lambda < 1).$$

$$v_{\infty, \lambda}^{\phi} = (I - \lambda P_{\phi})^{-1} r_{\phi}$$

$$\begin{aligned} v(s) &= \sum_{a \in \mathcal{A}} \phi(a|s) r(s, a) + \lambda \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \phi(a|s) P(s'|s, a) v(s') \\ &= \sum_{a \in \mathcal{A}} \phi(a|s) \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right] \end{aligned}$$

Question: How can we choose ϕ so that the corresponding $v_{\infty, \lambda}^{\phi}$ is maximum possible?

Definition: Given $\phi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$, define $L_{\phi}: \mathbb{R}^{|\mathcal{S}|} \rightarrow \mathbb{R}^{|\mathcal{S}|}$ as

$$\begin{aligned} L_{\phi} \underline{v} &= r_{\phi} + \lambda P_{\phi} \underline{v} \\ (L_{\phi} \underline{v})(s) &= \sum_{a \in \mathcal{A}} \phi(a|s) \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right]. \end{aligned}$$

Also, define $L: \mathbb{R}^{|\mathcal{S}|} \rightarrow \mathbb{R}^{|\mathcal{S}|}$ as the

$$(L \underline{v})(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right]$$

Remarks:

① From the above definitions, it is clear that

$$L \underline{v}(s) = \sup_{\phi} L_{\phi} \underline{v}(s) \quad \forall s \in \mathcal{S} \quad \forall \underline{v} \in \mathbb{R}^{|\mathcal{S}|}.$$

② The operator L is called the maximum return operator.

Definition (contraction mapping)

Let V be a vector space equipped with a norm $\|\cdot\|$.

An operator $T: V \rightarrow V$ is said to be a contraction mapping on V with modulus $\beta < 1$ if

$$\|T\underline{v} - T\underline{u}\| \leq \beta \|\underline{v} - \underline{u}\| \quad \forall \underline{u}, \underline{v} \in V.$$

Theorem 5.5 (Banach fixed point theorem / Contraction mapping theorem)

Let V be a complete, normed vector space (a.k.a Banach space).
Let $T: V \rightarrow V$ be a contraction. Then;

① There exists a unique $\underline{v}^* \in V$ such that

$$\underline{v}^* = T\underline{v}^*. \quad \leftarrow \text{fixed-point equation}$$

② Given any starting point $\underline{v}_0 \in V$, by applying the recursion

$$\underline{v}_k = T\underline{v}_{k-1}, \quad k \in \mathbb{N},$$

the sequence $\{\underline{v}_k\}_{k \geq 0}$ so generated converges to $\underline{v}^*$, i.e.,

$$\|\underline{v}_k - \underline{v}^*\| \xrightarrow{k \rightarrow \infty} 0.$$

Proof of Theorem 5.5 :

For any $m > n$, we have

$$\begin{aligned} \|\underline{v}_m - \underline{v}_n\| &\leq \sum_{k=n+1}^m \|\underline{v}_k - \underline{v}_{k-1}\| \\ &= \sum_{k=n+1}^m \|T\underline{v}_{k-1} - T\underline{v}_{k-2}\| \end{aligned}$$

$$\begin{aligned} &\leq \sum_{k=n+1}^m \beta \|\underline{v}_{k-1} - \underline{v}_{k-2}\| \\ &\vdots \\ &\leq \sum_{k=n+1}^m \beta^{k-1} \|\underline{v}_1 - \underline{v}_0\| \end{aligned}$$

$$\|\underline{v}_m - \underline{v}^*\| = \|T\underline{v}_{m-1} - T\underline{v}^*\|$$

$$\leq \beta \|\underline{v}_{m-1} - \underline{v}^*\|$$

$$\vdots$$

$$\leq \beta^n \|\underline{v}_0 - \underline{v}^*\|$$

$$\begin{aligned} &\leq \|\underline{v}_1 - \underline{v}_0\| \cdot \sum_{k=n+1}^{\infty} \beta^{k-1} \\ &= \|\underline{v}_1 - \underline{v}_0\| \cdot \frac{\beta^n}{1-\beta}. \end{aligned}$$

Letting $m, n \rightarrow \infty$, we get that

$$\|\underline{v}_m - \underline{v}_n\| \rightarrow 0 \quad \text{as } m, n \rightarrow \infty$$

$\Rightarrow \{\underline{v}_k\}_{k \geq 0}$ is a Cauchy sequence in V

$\Rightarrow \exists \underline{v}^* \in V$ such that $\underline{v}_k \xrightarrow[k \rightarrow \infty]{} \underline{v}^*$ because V is complete, every Cauchy sequence converges in V $\left(\|\underline{v}_k - \underline{v}^*\| \xrightarrow[k \rightarrow \infty]{} 0 \right)$

Lastly,

$$\begin{aligned} \|\underline{T}\underline{v}^* - \underline{v}^*\| &\leq \|\underline{T}\underline{v}^* - \underline{T}\underline{v}_n\| + \|\underline{T}\underline{v}_n - \underline{v}^*\| \\ &\leq \beta \|\underline{v}^* - \underline{v}_n\| + \|\underline{v}_{n+1} - \underline{v}^*\| \quad \forall n \in \mathbb{N}. \end{aligned}$$

Taking limits as $n \rightarrow \infty$, and using the fact that

$$\underline{v}_n \xrightarrow[n \rightarrow \infty]{} \underline{v}^*, \text{ we get}$$

$$\|\underline{T}\underline{v}^* - \underline{v}^*\| \leq 0 \Rightarrow \|\underline{T}\underline{v}^* - \underline{v}^*\| = 0 \Rightarrow \underline{T}\underline{v}^* = \underline{v}^*.$$

The below result shows that the operator L is a contraction.

Proposition 5.1:

Define $L: \mathbb{R}^{|\mathcal{S}|} \rightarrow \mathbb{R}^{|\mathcal{S}|}$ as

$$L\underline{v}(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right], \quad s \in \mathcal{S}.$$

Then, L is a contraction mapping on $\mathbb{R}^{|\mathcal{S}|}$ with modulus λ .

Proof of proposition 5.1

Fix $s \in \mathcal{S}$. Let $\underline{u}, \underline{v} \in \mathbb{R}^{|\mathcal{S}|}$. Without loss of generality, let $L\underline{v}(s) \geq L\underline{u}(s)$.

$$\text{Let } a_{\underline{v}}^*(s) = \arg \max_a \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right]$$

We then have

$$\begin{aligned} r(s, a_{\underline{v}}^*(s)) + \sum_{s' \in \mathcal{S}} P(s'|s, a_{\underline{v}}^*(s)) u(s') \\ \leq \max_{a \in \mathcal{A}} \left[r(s, a) + \sum_{s' \in \mathcal{S}} P(s'|s, a) u(s') \right] = L\underline{u}(s). \end{aligned}$$

It then follows that

$$0 \leq L\underline{v}(s) - L\underline{u}(s)$$

$$\leq r(s, a_v^*(s)) + \sum_{s' \in \mathcal{S}} \lambda P(s'|s, a_v^*(s)) v(s') - \left[r(s, a_u^*(s)) + \sum_{s' \in \mathcal{S}} \lambda P(s'|s, a_u^*(s)) u(s') \right]$$

$$= \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a_v^*(s)) (v(s') - u(s'))$$

$$\leq \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a_v^*(s)) \|v - u\|_\infty$$

$$= \lambda \|v - u\|_\infty.$$

We therefore get

$$\max_{s: L_v(s) \geq L_u(s)} L_v(s) - L_u(s) \leq \lambda \|v - u\|_\infty$$

Similarly, taking those $s \in \mathcal{S}$ for which $L_u(s) \geq L_v(s)$, we get

$$\max_{s: L_u(s) \geq L_v(s)} L_u(s) - L_v(s) \leq \lambda \|u - v\|_\infty.$$

Combining, we get

$$\|L_v - L_u\|_\infty \leq \lambda \|v - u\|_\infty.$$

Exercise:

Given any $\phi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$, show that $L_\phi: \mathbb{R}^{|\mathcal{S}|} \rightarrow \mathbb{R}^{|\mathcal{S}|}$ defined by

$$L_\phi v = r_\phi + \lambda P_\phi v$$

is a contraction mapping on $\mathbb{R}^{|\mathcal{S}|}$ with modulus λ .

What is the unique solution to $v = L_\phi v$? $v^* = v_{0,\lambda}^\phi = (I - \lambda P_\phi)^{-1} r_\phi$

Theorem 5.6: If $\lambda \in [0, 1)$, then the fixed point equation

$$L_v = v$$

has a unique solution $v^* \in \mathbb{R}^{|\mathcal{S}|}$.

The proof of Theorem 5.6 is a simple consequence of Theorem 5.5, Proposition 5.1, and the fact that $\mathbb{R}^{|\mathcal{S}|}$ is complete (and hence Banach).

Existence of an Optimal Value Function

Given $\lambda \in [0, 1)$, the optimal value function $V_{\infty, \lambda}^* : \mathcal{S} \rightarrow \mathbb{R}$ is defined as

$$V_{\infty, \lambda}^*(s) := \sup_{\pi \in \Pi^{\text{MR}}} V_{\infty, \lambda}^{\pi}(s) \stackrel{\text{Theorem 5.1}}{=} \sup_{\pi \in \Pi^{\text{MR}}} V_{\infty, \lambda}^{\pi}(s), \quad s \in \mathcal{S}.$$

Theorem 5.7: Fix $\lambda \in [0, 1)$.

① If there exists $\underline{v} \in \mathbb{R}^{|\mathcal{S}|}$ such that

$$\underline{v} \geq L\underline{v} \quad (\text{component-wise}),$$

then $\underline{v} \geq V_{\infty, \lambda}^*$ (Component-wise).

② If there exists $\underline{v} \in \mathbb{R}^{|\mathcal{S}|}$ such that

$$\underline{v} \leq L\underline{v} \quad (\text{component-wise}),$$

then $\underline{v} \leq V_{\infty, \lambda}^*$ (Component-wise).

$$\sup_y f(y) \leq \alpha$$

③ If there exists $\underline{v} \in \mathbb{R}^{|\mathcal{S}|}$ such that

$$\underline{v} = L\underline{v} \quad (\text{component-wise}),$$

then $\underline{v} = V_{\infty, \lambda}^*$ (Component-wise).

$$\Leftrightarrow f(x) \leq \alpha \quad \forall x$$

Proof of Theorem 5.7:

Fix an arbitrary $\pi = \{\pi_0, \pi_1, \dots\} \in \Pi^{\text{MR}}$.

① For any $\phi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$, we have

$$\underline{v} \geq L\underline{v} = \sup_{\phi : \mathcal{S} \rightarrow \Delta(\mathcal{A})} L_{\phi} \underline{v} \geq L_{\phi} \underline{v}.$$

From this, it follows that

$$\begin{aligned} \underline{v} &\geq L_{\pi_0} \underline{v} = r_{\pi_0} + \lambda P_{\pi_0} \underline{v} \\ &\geq r_{\pi_0} + \lambda P_{\pi_0} (L_{\pi_1} \underline{v}) \quad (\because \underline{v} \geq L_{\pi_1} \underline{v}) \\ &= r_{\pi_0} + \lambda P_{\pi_0} [r_{\pi_1} + \lambda P_{\pi_1} \underline{v}] \\ &= r_{\pi_0} + \lambda P_{\pi_0} r_{\pi_1} + \lambda^2 P_{\pi_0} P_{\pi_1} \underline{v} \\ &\vdots \\ &= r_{\pi_0} + \lambda P_{\pi_0} r_{\pi_1} + \lambda^2 P_{\pi_0} P_{\pi_1} r_{\pi_2} + \dots \\ &= v_{\infty, \lambda}^{\pi} \end{aligned}$$

Because the above holds for any $\pi \in \Pi^{MR}$, taking $\sup_{\pi \in \Pi^{MR}}$ on both sides, we get

$$\underline{V} \geq \sup_{\tau \in \Pi_{MR}} V_{\infty, \tau}^{\pi} = V_{\infty, \tau}^{*}.$$

② We have

$$\underline{v} \leq L \underline{v} = \sup_{\phi} L_{\phi} \underline{v}$$

$$= \sup_{\phi} \sum_{a \in A} \phi(a|s) \left[r(s,a) + \lambda \sum_{s' \in S} P(s'|s,a) v(s') \right]$$

Note that $\exists \phi^* : \mathcal{S} \rightarrow \Delta(A)$ for which the sup is attained

Then,

$$\begin{aligned}
 \underline{v} &\leq L_{\phi^*} \underline{v} = \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right] \\
 &= r_{\phi^*} + \lambda P_{\phi^*} \underline{v} \quad \quad \quad := Q(s, a) \\
 &\leq r_{\phi^*} + \lambda P_{\phi^*} (L_{\phi^*} \underline{v}) \\
 &\leq r_{\phi^*} + \lambda P_{\phi^*} (r_{\phi^*} + \lambda P_{\phi^*} \underline{v}) \\
 &= r_{\phi^*} + \lambda P_{\phi^*} r_{\phi^*} + \lambda^2 P_{\phi^*}^2 \underline{v} \\
 &\vdots
 \end{aligned}$$

$$\leq r_{\phi^*} + \lambda P_{\phi^*} r_{\phi^*} + \lambda^2 P_{\phi^*}^2 r_{\phi^*} + \dots$$

$$= V_{\infty, \lambda}^{\phi^*} \quad (\text{corresponding to } \pi = \{\phi^*, \phi^*, \dots\})$$

$$\leq \sup_{\pi \in \Pi^{\text{HR}}} V_{\infty, \lambda}^{\pi} = V_{\infty, \lambda}^*.$$

③ Follows from ① and ②.

Theorem 5.8 (On the existence of optimal value function)

Fix $\lambda \in [0, 1)$. The optimal value function $V_{\infty, \lambda}^*$ is the unique solution to the fixed point equation

$$\underline{V} = L \underline{V}.$$

Proof of Theorem 5.8:

From Proposition 5.1 and Theorem 5.6, we know that L is a contraction and the equation $\underline{V} = L \underline{V}$ has a unique fixed point $\underline{V}^*$.

From Theorem 5.7, we know that $V^* = V_{\infty, \lambda}^*$.

Existence of an Optimal Stationary Policy

Theorem 5.9: Let $\phi^* : \mathcal{S} \rightarrow \mathcal{A}$ be defined as

$$\phi^*(s) = \arg \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) V_{\infty, \lambda}^*(s') \right], \quad s \in \mathcal{S}.$$

Then, ϕ^* is an optimal policy, i.e., $V_{\infty, \lambda}^{\phi^*} = V_{\infty, \lambda}^*$.

Proof of Theorem 5.9:

We have

Theorem 5.8

$$V_{\infty, \lambda}^* \stackrel{\downarrow}{=} L_{\infty, \lambda} V_{\infty, \lambda}^*$$

$$= \sup_{\phi} L_{\phi} V_{\infty, \lambda}^*$$

$$= \sup_{\phi} \sum_{a \in \mathcal{A}} \phi(a|s) \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) V_{\infty, \lambda}^*(s') \right]$$

$$= \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) V_{\infty, \lambda}^*(s') \right]$$

$$= \left[r(s, \phi^*(s)) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, \phi^*(s)) V_{\infty, \lambda}^*(s') \right]$$

$$= L_{\phi^*} V_{\infty, \lambda}^*$$

From the exercise, we know that L_{ϕ^*} is a contraction, and the unique solution to $\underline{v} = L_{\phi^*} \underline{v}$ is

$$V_{\infty, \lambda}^{\phi^*} = (\mathbf{I} - \lambda P_{\phi^*})^{-1} r_{\phi^*}.$$

From this, it follows that

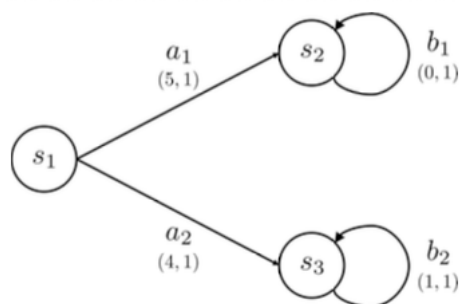
$$V_{\infty, \lambda}^* = V_{\infty, \lambda}^{\phi^*} = (\mathbf{I} - \lambda P_{\phi^*})^{-1} r_{\phi^*}.$$

Corollary: The policy $\pi^* = \{\phi^*, \phi^*, \dots\} \in \Pi^{\text{SD}}$ is optimal.

Remarks:

- ① The optimal policy π^* is called a greedy policy.
- ② The optimal policy π^* may depend on λ , in general. ↖ see example below
- ③ The fixed-point equation $\underline{v} = L \underline{v}$ is called the Bellman equation.

Example :



Fix a discount factor
 $\lambda \in [0, 1)$.

$$V_{\infty, \lambda}^*(s_1) = \max \left\{ \overbrace{5 + \lambda V_{\infty, \lambda}^*(s_2)}^{a_1}, \overbrace{4 + \lambda V_{\infty, \lambda}^*(s_3)}^{a_2} \right\}$$

$$V_{\infty, \lambda}^*(s_2) = 0 + \lambda V_{\infty, \lambda}^*(s_2)$$

$$V_{\infty, \lambda}^*(s_3) = 1 + \lambda V_{\infty, \lambda}^*(s_3).$$

The optimal value function is then given by

$$V_{\infty, \lambda}^*(s) = \begin{cases} \max \left\{ 5, 4 + \frac{\lambda}{1-\lambda} \right\}, & s = s_1, \\ 0, & s = s_2, \\ \frac{1}{1-\lambda}, & s = s_3. \end{cases}$$

The optimal $\phi^* : \mathcal{S} \rightarrow \mathcal{A}$ is such that

$$\phi^*(s_1) = \begin{cases} a_1, & 5 \geq 4 + \frac{\lambda}{1-\lambda}, \\ a_2, & \text{otherwise.} \end{cases}$$
